

Case Study – 3: RetailMax Electronics - Business Analytics Challenge

Background

You are part of the Business Analytics Team at **RetailMax Electronics**, a growing electronics retail chain in India. The company operates across 5 major cities with both online and offline presence:

Company Structure:

- **Mumbai** – 3 stores + online hub (largest market)
- **Delhi** – 2 stores + online operations
- **Bangalore** – 2 stores + strong online presence
- **Chennai** – 2 stores + emerging online market
- **Pune** – 1 store + online operations

Product Portfolio:

- **Smartphones** – 40% of revenue (high competition)
- **Laptops** – 35% of revenue (good margins)
- **Tablets** – 15% of revenue (declining trend)
- **Accessories** – 10% of revenue (highest profit margins)

The Problem (Critical Business Issues)

RetailMax is facing declining profitability despite growing sales volumes!

Key Challenges:

1. **Inconsistent Performance** – Some cities/channels performing much better than others
2. **Profit Margin Pressure** – Average profit margins dropped from 28% to 22% in Q4 2024
3. **Inventory Issues** – Overstocking slow-moving products, understocking popular items
4. **Sales Team Efficiency** – Huge variation in individual sales rep performance
5. **Customer Retention** – Need better insights on new vs existing customer behavior

Critical Questions Management Needs Answered:

- Which cities/channels should get more investment?
- Which products are actually profitable vs just high-volume?
- How to optimize sales team performance and commission structure?
- What's the real customer lifetime value pattern?

Your Mission

Help RetailMax optimize operations and boost profitability using Excel analytics!

TASK 1: Basic Analysis & Formatting

Apply Conditional Formatting:

1. Highlight profit margins >25% in GREEN
2. Highlight profit margins <15% in RED
3. Add data bars to Total_Sales column

Create Reference Tables: Create a separate area with:

- Sales Rep Commission Table:
 - Sales <₹50,000 = 2% commission
 - Sales ≥₹50,000 = 3% commission
- Use VLOOKUP to add commission amount for each transaction

TASK 2: Pivot Table Analysis

1. **Sales by City and Channel**
 - Rows: City, Channel
 - Values: Sum of Total_Sales, Count of Transaction_ID
 - Add calculated field for Total Profit
2. **Monthly Performance**
 - Rows: Month (group dates by month)
 - Columns: Category
 - Values: Sum of Total_Sales
3. **Product Category Analysis**
 - Rows: Category, Product_Name
 - Values: Sum of Total_Sales, Sum of Quantity
 - Filter by City
4. **Customer Type Analysis**
 - Rows: Customer_Type, Payment_Method
 - Values: Average of Total_Sales, Count of transactions

TASK 3: Dashboard Creation

Create a Business Dashboard including:

1. Top 5 performing cities (use formulas)
2. Best-selling product categories (from pivot data)
3. Monthly sales trend chart

Deliverables:

1. **Complete Excel File** with all sheets properly formatted
2. **Business Insights Summary** (in a text box or separate sheet)
3. What recommendations would you give for Q1 2025 strategy?
4. **Top 3 Strategic Recommendations** based on your analysis